# Lecture 13 recap

In Lecture 13:

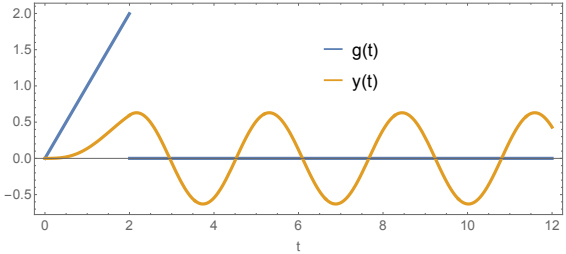
- Went through examples on using LTs to solve IVPs with discontinuous forcing
  - Looked at off–on forcing and how this produces discontinuities in a solution derivative
  - Learnt how to represent a ramp-like forcing using Heaviside step functions

# Lecture 13 recap

In Lecture 13:

- Went through examples on using LTs to solve IVPs with discontinuous forcing
  - Looked at off–on forcing and how this produces discontinuities in a solution derivative
  - Learnt how to represent a ramp-like forcing using Heaviside step functions

$$y(t) = \frac{1}{4}t - \frac{1}{8}\sin 2t - \left(\frac{t}{4} - \frac{1}{2}\cos 2(t-2) - \frac{1}{8}\sin 2(t-2)\right) H(t-2)$$



# Lecture 14

In Lecture 14, we will:

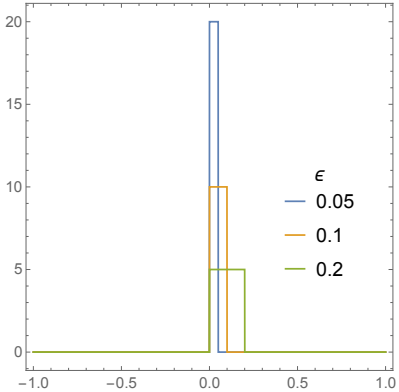
1. Introduce the Dirac delta and use it to model impulsive forcing
2. Introduce convolutions as a tool to calculate ILTs

# Impulses

Suppose we want to model a force acting over a negligibly small time period, i.e., an impulse

Use a rectangular pulse of unit area:

$$\delta_\epsilon(t) = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon \\ 0 & t > \epsilon \end{cases}, \quad \text{“small” } \epsilon > 0$$

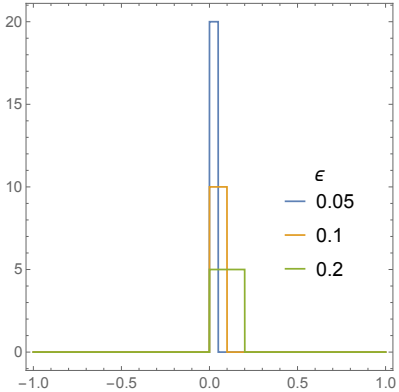


# Impulses

Suppose we want to model a force acting over a negligibly small time period, i.e., an impulse

Use a rectangular pulse of unit area:

$$\delta_\epsilon(t) = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon \\ 0 & t > \epsilon \end{cases}, \quad \text{“small” } \epsilon > 0$$



# Impulses

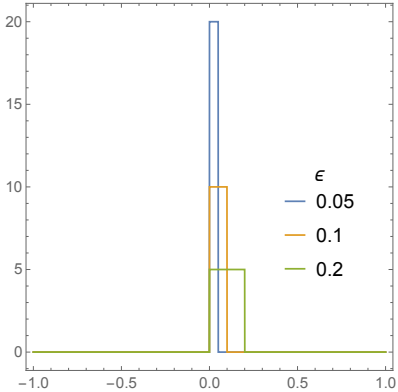
Suppose we want to model a force acting over a negligibly small time period, i.e., an impulse

Use a rectangular pulse of unit area:

$$\delta_\epsilon(t) = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon \\ 0 & t > \epsilon \end{cases}, \qquad \text{“small” } \epsilon > 0$$

The ‘strength’ of this impulse is

$$\int_0^\infty \delta_\epsilon(t)\,\mathrm{d}t = \int_0^\epsilon \frac{1}{\epsilon}\,\mathrm{d}t = 1$$




As width $\to 0$, height $\to \infty$
but area remains 1

# Limit as $\epsilon \to 0$

1. $\lim_{\epsilon \to 0} \delta_\epsilon = 0$ for all $t \neq 0$

# Limit as $\epsilon \to 0$

1. $\lim_{\epsilon \to 0} \delta_\epsilon = 0$ for all $t \neq 0$
2. $\lim_{\epsilon \to 0} \int_0^\infty \delta_\epsilon \, \mathrm{d}t = 1$

*No function (in the traditional sense) can have these properties!*

# Limit as $\epsilon \to 0$

$$\mathcal{L}\{\delta_\varepsilon\} = \int_0^\infty e^{-st}\,\delta_\varepsilon(t)\,dt$$

$$= \frac{1}{\varepsilon}\int_0^\varepsilon e^{-st}\,dt$$

$$= \frac{-1}{\varepsilon s}\left[e^{-st}\right]_0^\varepsilon$$

$$= \frac{1}{\varepsilon s}\left(1 - e^{-\varepsilon s}\right)$$

1. $\lim_{\epsilon\to 0} \delta_\epsilon = 0$ for all $t \neq 0$
2. $\lim_{\epsilon\to 0} \int_0^\infty \delta_\epsilon \, \mathrm{d}t = 1$
3. $\lim_{\epsilon\to 0} \mathscr{L}(\delta_\epsilon) = \lim_{\epsilon\to 0} \dfrac{1 - \mathrm{e}^{-\epsilon s}}{\epsilon s} = 1$ (L'Hôpital's rule)

*No function (in the traditional sense) can have these properties!*

# The Dirac delta, $\delta(t)$

## Definition

We define $\delta(t)$ such that

1. $\delta(t) = 0$ for all $t \neq 0$,
2. $\displaystyle\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = \int_{0}^{a} \delta(t)\mathrm{d}t = 1$ for all $a > 0$, and
3. $\mathscr{L}\{\delta(t)\} = 1$.

- $\delta(t)$ is not a function in the traditional sense (it is a *generalised function* as you will discover in later courses)
- There are various forms of $\delta(t)$ with others being centred at or left of zero; we will always use the above definition

# Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

# Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

- Let $c \geqslant 0$, then

# Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

- Let $c \geqslant 0$, then

1. $\displaystyle\int_0^\infty f(t)\delta(t-c)\ \mathrm{d}t = f(c)$

# Sifting property

- Since $\delta(t)$ is zero for $t \neq 0$, it 'sifts out' the value of a function at $t = 0$

$$\int_0^\infty f(t)\delta(t)\mathrm{d}t = f(0)$$

- Let $c \geqslant 0$, then

$$\mathcal{L}^{-1}\{e^{-sc}\} = \delta(t-c)$$

1. $\displaystyle\int_0^\infty f(t)\delta(t-c)\ \mathrm{d}t = f(c)$

2. $\mathscr{L}\Big\{\delta(t-c)\Big\} = \displaystyle\int_0^\infty \mathrm{e}^{-st}\delta(t-c)\ \mathrm{d}t = \mathrm{e}^{-sc}$

## Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

## Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

# **Exercise 3.19**: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

$$\mathscr{L}\{y'\} - \mathscr{L}\{y\} = sY - \cancel{y(0)} - Y$$

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

$$\Rightarrow \quad sY - Y = \mathrm{e}^{-s}$$

## Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

$$\Rightarrow \quad sY - Y = \mathrm{e}^{-s}$$

$$Y(s) = \frac{\mathrm{e}^{-s}}{s-1}$$

## Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

$$\Rightarrow \quad sY - Y = \mathrm{e}^{-s}$$

$$Y(s) = \frac{\mathrm{e}^{-s}}{s-1}$$

$$y(t) = \mathscr{L}^{-1}\left(\frac{\mathrm{e}^{-s}}{s-1}\right)$$

# Exercise 3.19: Solve $y'(t) - y(t) = \delta(t-1)$ where $y(0) = 0$

We solve the problem using Laplace transforms.

$$\mathscr{L}\{y' - y\} = \mathscr{L}\{\delta(t-1)\}$$

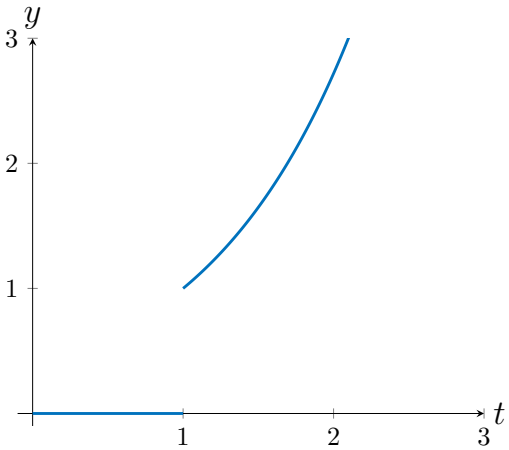
$$\Rightarrow \quad sY - Y = \mathrm{e}^{-s}$$

$$Y(s) = \frac{\mathrm{e}^{-s}}{s-1}$$

$$y(t) = \mathscr{L}^{-1}\left(\frac{\mathrm{e}^{-s}}{s-1}\right) = \mathrm{e}^{t-1}H(t-1) \quad t\text{-shifting}$$

$$\mathscr{L}^{-1}\{1/s-1\} = e^t$$

$$y(t) = \mathrm{e}^{t-1} H(t-1)$$

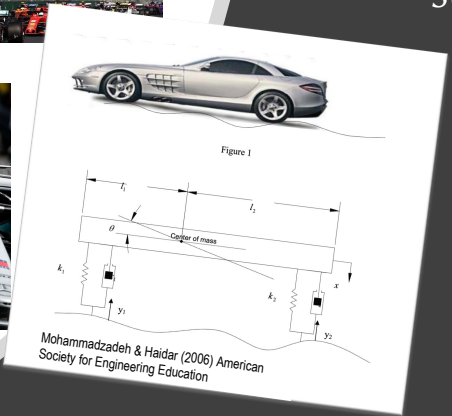



The solution satisfies the initial condition and the ODE but it is discontinuous

# Case study 4: Formula 1 car suspension

https://www.planetf1.com/news/sydney-australian-grand-prix-bid/

https://www.mercedesamgf1.com/en/news/2019/06/the-suspension-of-a-formula-one-car/




- Read all about formula 1 car suspensions here https://www.mercedesamgf1.com/en/news/2019/06/the-suspension-of-a-formula-one-car/
- **Mathematical models lead to ODEs which can be solved using Laplace transforms.**

# Convolution

## Definition (Convolution)

Given two functions, $f$ and $g$, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$f * g = \int_0^t f(t-\tau)g(\tau)\mathrm{d}\tau$$

# Convolution

## Definition (Convolution)

Given two functions, $f$ and $g$, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$f * g = \int_0^t f(t-\tau)g(\tau)\mathrm{d}\tau$$

## Basic properties

- Convolutions are commutative, i.e., $f * g = g * f$

# Convolution

## Definition (Convolution)

Given two functions, $f$ and $g$, defined on $[0, \infty)$, their convolution, $f * g$ is defined

$$f * g = \int_0^t f(t-\tau)g(\tau)\mathrm{d}\tau$$

## Basic properties

- Convolutions are commutative, i.e., $f * g = g * f$
- In general, $1 * f \neq f$, as

$$(1 * f)(t) = \int_0^t f(\tau)\mathrm{d}\tau, \quad \text{i.e., an antiderivative}$$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

$$\underbrace{\mathrm{e}^t}_{f(t)} * \underbrace{\mathrm{e}^{2t}}_{g(t)} = \int_0^t \overbrace{\mathrm{e}^{t-\tau}}^{f(t-\tau)} \overbrace{\mathrm{e}^{2\tau}}^{g(\tau)} \, \mathrm{d}\tau$$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

$$\begin{aligned}\mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\ &= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau\end{aligned}$$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

$$
\begin{aligned}
\mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\
&= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau \\
&= \mathrm{e}^t(\mathrm{e}^t - 1)
\end{aligned}
$$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

$$\begin{aligned}
\mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\
&= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau \\
&= \mathrm{e}^t(\mathrm{e}^t - 1)
\end{aligned}$$

Note that

$$\mathrm{e}^{2t} * \mathrm{e}^t = \int_0^t \mathrm{e}^{2(t-\tau)}\mathrm{e}^{\tau}\ \mathrm{d}\tau$$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

$$\begin{aligned}
\mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\
&= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau \\
&= \mathrm{e}^t(\mathrm{e}^t - 1)
\end{aligned}$$

Note that

$$\mathrm{e}^{2t} * \mathrm{e}^t = \int_0^t \mathrm{e}^{2(t-\tau)}\mathrm{e}^{\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}\int_0^t \mathrm{e}^{-\tau}\ \mathrm{d}\tau$$

## **Exercise 3.20**: Calculate the convolution of $\mathrm{e}^t$ with $\mathrm{e}^{2t}$

$$\begin{aligned}
\mathrm{e}^t * \mathrm{e}^{2t} &= \int_0^t \mathrm{e}^{t-\tau}\mathrm{e}^{2\tau}\ \mathrm{d}\tau \\
&= \mathrm{e}^t \int_0^t \mathrm{e}^{\tau}\mathrm{d}\tau \\
&= \mathrm{e}^t(\mathrm{e}^t - 1)
\end{aligned}$$

Note that

$$\mathrm{e}^{2t} * \mathrm{e}^t = \int_0^t \mathrm{e}^{2(t-\tau)}\mathrm{e}^{\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}\int_0^t \mathrm{e}^{-\tau}\ \mathrm{d}\tau = \mathrm{e}^{2t}(1-\mathrm{e}^{-t}) = \mathrm{e}^t(\mathrm{e}^t - 1)$$

**Exercise 3.21**: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

## **Exercise 3.21**: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

### Laplace transform of the convolution

Let $f$ and $g$ be functions with Laplace transforms, then

$$\mathscr{L}(f * g) = F(s)G(s)$$

*Laplace transforms convert convolutions into multiplication*

## Exercise 3.21: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

$F(s)G(s)$

$F(s) = \frac{1}{s-1}$ ; $G(s) = \frac{1}{s-2}$

### Laplace transform of the convolution

Let $f$ and $g$ be functions with Laplace transforms, then

$$\mathscr{L}(f * g) = F(s)G(s)$$

$\mathcal{L}^{-1}\{FG\} = f * g$

*Laplace transforms convert convolutions into multiplication*

$$\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} = \mathrm{e}^{t} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\} = \mathrm{e}^{2t}$$

$e^{t} * e^{2t}$

# **Exercise 3.21**: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

## Laplace transform of the convolution

Let $f$ and $g$ be functions with Laplace transforms, then

$$\mathscr{L}(f * g) = F(s)G(s)$$

*Laplace transforms convert convolutions into multiplication*

$$\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} = \mathrm{e}^{t} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\} = \mathrm{e}^{2t}$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{1}{s-1}\frac{1}{s-2}\right\} = \mathrm{e}^{t} * \mathrm{e}^{2t}$$

# **Exercise 3.21**: Calculate $\mathscr{L}^{-1}\left\{\dfrac{1}{(s-1)(s-2)}\right\}$

## Laplace transform of the convolution

Let $f$ and $g$ be functions with Laplace transforms, then

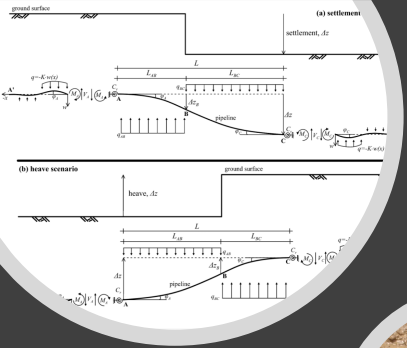
$$\mathscr{L}(f * g) = F(s)G(s)$$

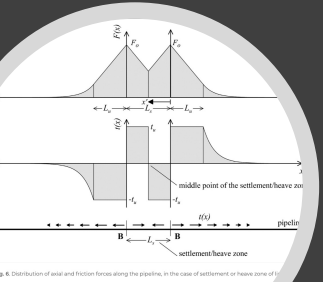
*Laplace transforms convert convolutions into multiplication*

$$\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} = \mathrm{e}^{t} \quad \text{and} \quad \mathscr{L}^{-1}\left\{\frac{1}{s-2}\right\} = \mathrm{e}^{2t}$$

$$\Rightarrow \quad \mathscr{L}^{-1}\left\{\frac{1}{s-1}\frac{1}{s-2}\right\} = \mathrm{e}^{t} * \mathrm{e}^{2t}$$

From previous exercise $\mathrm{e}^{t} * \mathrm{e}^{2t} = \mathrm{e}^{t}(\mathrm{e}^{t} - 1)$






g. 6. Distribution of axial and friction forces along the pipeline, in the case of settlement or heave zone of l

# Case study 5: Critical pipeline performance

- Buried pipelines are critical lifeline systems, supplying drinking water and natural gas to communities. Damage to these pipelines caused by ground deformation and related hazards like earthquakes and landslides can lead to severe socioeconomic losses to unprepared communities.
- Mitigation measures employ mathematical models of these structures and these often involve ordinary differential equations with discontinuous forcing and impulsive functions
- Images from Lee et al (2016) Quarterly Journal of Engineering Geology and Hydrogeology, and Kouretzis et al (2015) Canadian Geotechnical Journal

## **Exercise 3.22**: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

## Exercise 3.22: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

**Laplace transform of an antiderivative**

$$\mathscr{L}\left(\int_0^t f(\tau)\mathrm{d}\tau\right) = \mathscr{L}(1 * f) = \mathscr{L}(1)F(s) = \frac{F(s)}{s}$$

$$\mathcal{L}\{y'\} + 4\mathcal{L}\left\{\int_0^t y(\tau)d\tau\right\} = 0$$

$$sY - \cancel{y(0)}^{3} + 4\,Y/s = 0$$

$$s^2Y - 3s + 4Y = 0 \Rightarrow \{s^2+4\}Y = 3s/s^2+4$$

$$\therefore\ y = \mathcal{L}^{-1}\{Y\} = 3\mathcal{L}^{-1}\{s/s^2+2^2\} = 3\cos 2t$$

## **Exercise 3.22**: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

**Laplace transform of an antiderivative**

$$\mathscr{L}\left(\int_0^t f(\tau)\mathrm{d}\tau\right) = \mathscr{L}(1 * f) = \mathscr{L}(1)F(s) = \frac{F(s)}{s}$$

Taking the Laplace transform gives

$$\begin{aligned}
sY - 3 + 4Y * \mathscr{L}(1) = 0 \qquad &\Rightarrow \qquad sY - 3 + \frac{4Y}{s} = 0 \\
s^2Y - 3s + 4Y = 0 \qquad &\Rightarrow \qquad (s^2+4)Y = 3s \\
Y = \frac{3s}{s^2+4} \qquad &\Rightarrow \qquad Y = \mathscr{L}^{-1}\left(\frac{3s}{s^2+4}\right) = 3\cos 2t
\end{aligned}$$